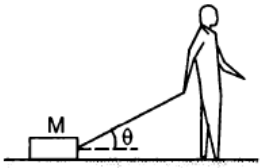
PHY101: Introduction to Physics-1

Monsoon, 2017

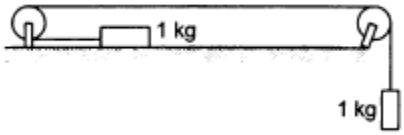
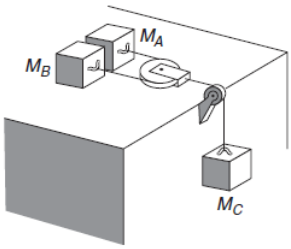
Tutorial-5

1. A block of mass M is pulled on a smooth horizontal table by a string making an angle θ with the horizontal as shown in the figure below. If the acceleration of the block is *a*, find the force applied by the string and by the table on the block.



2. At $t = 0$, an elevator departs from the ground with uniform speed. At time T_1 a child drops a marble through the floor. The marble falls with uniform acceleration $g = 9.8$ m/s^2, and hits the ground T_2 seconds later. Find the height of the elevator at time T_1.

3. Calculate the tension in the string shown in figure below. The pulley and the string are light and all surfaces are frictionless. Take $g = 10$ m/s^2.



4. Two masses, M_A and M_B, lie on a frictionless table, as shown. They are attached to either end of a light rope of length *l* which passes around a pulley of negligible mass. The pulley is attached to a rope connected to a hanging mass, M_C. Find the acceleration of each mass.